

The Zoppoli Research Underground — Station & Connection Guide

How to ride each line: what every station does, **how to connect it**, **which functions to expect**, whether Claude reaches it **automatically or via a skill load**, and the **machine it runs best on**.

Generated from the live map data — **78 stations** (the super-confident set: ● live + ● verified). Registry-listed servers stay in `CATALOGUE.md` until their repo is confirmed. Re-run `python3 make_guide.py` after editing the map.

On the `Real?` column (is it actually in there, or did I make it up): the live MCPs and skills are ones I *directly observe* in this session; the public databases are well-known resources; and every install-first MCP repo / endpoint was **fetchd and confirmed to exist on 2026-06-04** (licence noted). Nothing here is an unverified guess — the registry-listed servers that *were* only inferred are excluded (they live in `CATALOGUE.md`).

How to read the columns

Status: ● live (wired now) · ● verified (real, ready to wire) · ⊕ interchange (a connecting line stops here).

Invocation — how Claude actually reaches it:

Label	Meaning
<code>Auto·MCP</code>	Wired MCP tool — Claude calls it directly; the tool schema auto-loads on first use. No action from you.
<code>Auto·web</code>	Public API/registry — Claude reaches it through the in-browser tool or <code>curl</code> . No install.
<code>Auto·Bash</code>	Claude runs it as local code / CLI in the shell.
<code>Skill</code>	Claude loads a skill (auto-triggers on the matching task, or on request).
<code>Install→MCP</code>	Add the server to your MCP config once ; thereafter it behaves as <code>Auto·MCP</code> .
<code>Install→Bash</code>	Install the tool once (<code>pip</code> / <code>conda</code>), then <code>Auto·Bash</code> .
<code>Install→Skill</code>	<code>/plugin install ...</code> , then it loads like any skill.
<code>Install (channel)</code>	An install <i>source</i> (the marketplace), not a tool you call.
<code>Built-in</code>	Native Claude Code capability — no setup.

Label	Meaning
Manual	You run the final step yourself (e.g. paste a prompt into ChatGPT).
Discovery	A place to find/verify other servers (browse), not a tool you call.

Runs best on: 🖥️ any laptop (cloud / remote MCP / API / skill — the work happens off-box) · 🏠 your workstation (local envs, auth tokens, your data, light–medium compute) · 🏢 HPC (heavy pipelines, big data, GPU). **A → B** = starts on A, scales to B.

Cost: **Free·OSS** open-source software/MCP · **Free·public** public DB/API · **Free·skill** a Claude skill (no licence; it runs on your Claude plan) · **Freemium** free tier + paid upgrades · **Paid/lic.** needs a subscription or institutional licence. Parentheticals flag where the answer shifts (e.g. image-gen skills bill per image; OncoKB is free for academics, licensed for industry).

Real? — provenance, the honest part: **Seen·session** I directly observe it in my tools/skills right now (cannot be fabricated) · **Known·DB** a well-established public database · **Known·SW** well-established open-source software (the *installed-on-your-box* claim is from a prior session) · **Web-verified** ✓ I fetched the repo/endpoint on 2026-06-04 and confirmed it exists, licence noted · **Configured** set up per CLAUDE.md, not exercised this turn.

Wire these first (highest-leverage, not yet wired)

1.

BioMCP — One server federating literature, variants, trials & cancer genomics. *(MCP: genomoncology/biomcp — the single best biomedical addition.)*
2.

Open Targets MCP — Target–disease–drug associations, genetics, tractability. *(Official remote MCP: mcp.platform.opentargets.org/mcp.)*
3.

cBioPortal MCP — No-SQL natural-language queries over cBioPortal. *(Official MCP (under construction): cBioPortal/cbioportal-mcp.)*
4.

ChatSpatial — Spatial-transcriptomics agent — ~60 methods (Xenium/Visium/MERFISH). *(MCP: cafferychen777/ChatSpatial.)*
5.

SCMCP — scRNA-seq analysis via scverse (scanpy). *(MCP: scmcpclub/scmcp.)*
6.

ClinTrials.gov MCP — Trial-registry queries over MCP (CTG API v2 + NCI CTS). *(MCP: cyanheads/clinicaltrials.gov-mcp-server.)*
7.

ENA MCP — European Nucleotide Archive — reads, assemblies, sequence records. *(MCP: biocontext-ai/nucleotide_archive_mcp.)*
8.

Claude for Life Sciences — Anthropic’s official life-sci plugin marketplace — remote MCPs + skills. *(Install: /plugin marketplace add anthropics/life-sciences.)*

🔴 Literature & Evidence

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
●🕒 PubMed	35M+ biomedical citations; full-text, related-article & citation lookup. ↳ MCP connected now — search, fetch metadata & full text directly.	Free·public	Seen·session	Auto·MCP	🖥️

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Consensus	Evidence synthesis with claim-level support/oppose scoring. ↳ MCP connected now — ask a question, get cited consensus.	Freemium	Seen-session	Auto·MCP	
● deep-research	Fan-out web search → fetch → adversarially verify → cited report. ↳ Skill: deep-research.	Free-skill	Seen-session	Skill	
● literature-review	Systematic reviews across PubMed/arXiv/bioRxiv/Semantic Scholar (PRISMA). ↳ Skill: claude-scientific-writer:literature-review.	Free-skill	Seen-session	Skill	
● research-lookup	Fast current-research lookup (Parallel/Perplexity) for facts & papers. ↳ Skill: claude-scientific-writer:research-lookup.	Free-skill (API backend may bill)	Seen-session	Skill	
● Europe PMC	EU biomedical corpus incl. preprints, patents, grants, full text. ↳ Public REST API via browser/HTTP; a Europe PMC MCP can be added.	Free-public	Known-DB	Auto·web	
● bioRxiv	Biology preprints — earliest signal before peer review. ↳ bioRxiv API via browser/HTTP.	Free-public	Known-DB	Auto·web	
● medRxiv	Clinical/medical preprints — trials & methods before journals. ↳ medRxiv API via browser/HTTP.	Free-public	Known-DB	Auto·web	
● Semantic Scholar	200M+ papers, citation graph, TLDR summaries. ↳ Public Graph API; also federated inside BioMCP.	Free-public	Known-DB	Auto·web	

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Wiley Scholar Gateway	Wiley full-text scholarly gateway (marketplace remote MCP). ↳ Install via /plugin marketplace add anthropics/life-sciences.	Paid/lic. (Wiley institutional)	Web-verified ✓ (in marketplace)	Install→MCP	

● Genomic Data & Sequencing

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Synapse.org	Sage Bionetworks repository — controlled-access cohorts, consortium data. ↳ MCP connected now — search & pull datasets (auth for controlled access).	Free-public (access-gated)	Seen-session	Auto·MCP	 query ·  bulk
● GEO / SRA	NCBI expression (GEO) & raw reads (SRA) — public omics at scale. ↳ NCBI E-utilities API; or an NCBI MCP (see below).	Free-public	Known-DB	Auto·web	 meta ·  raw reads
● Ensembl	Genome annotation, BioMart, VEP variant effect prediction. ↳ Ensembl REST API; or an Ensembl MCP (see below).	Free-public	Known-DB	Auto·web	
● UCSC Browser	Genome browser, tracks, liftOver, table browser. ↳ UCSC REST API; browser for exploration.	Free-public	Known-DB	Auto·web	
● GDC / TCGA	NCI Genomic Data Commons + TCGA — mutations, CNV, expression, /analysis/survival. ↳ Public REST API (gdc.cancer.gov/developers); no GDC MCP yet.	Free-public	Known-DB	Auto·web	 ·  bulk files

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
🕒🕒 ClinVar / gnomAD	Clinical variant significance + population allele frequencies. ↳ NCBI / gnomAD APIs; also federated inside BioMCP.	Free-public	Known-DB	Auto·web	🖥️
🕒🕒 cBioPortal	Cancer genomics portal — TCGA, mutations, CNA, clinical. ↳ REST API (cbioportal.org/api); MCP below.	Free-public	Known-DB	Auto·web	🖥️
🕒🕒 DepMap / OncoKB	Dependency/essentiality screens + precision-oncology annotation. ↳ DepMap downloads; OncoKB API (token).	Free-public (OncoKB: free academic / paid commercial)	Known-DB	Auto·web	🖥️
🕒 CELLxGENE	Single-cell/spatial atlases — directly relevant to your Xenium work. ↳ CZ CELLxGENE Census API; Census MCP below.	Free-public	Known-DB	Auto·Bash	🖥️ → 📄
🕒🕒 cBioPortal MCP	No-SQL natural-language queries over cBioPortal. ↳ Official MCP (under construction): cBioPortal/cbioportal-mcp.	Free-OSS	Web-verified ✓ MIT	Install→MCP	🖥️
🕒🕒 Open Targets MCP	Target–disease–drug associations, genetics, tractability. ↳ Official remote MCP: mcp.platform.opentargets.org/mcp.	Free-OSS	Web-verified ✓ official, Apache-2.0	Install→MCP	🖥️ (remote)
🕒 ENA MCP	European Nucleotide Archive — reads, assemblies, sequence records. ↳ MCP: biocontext-ai/nucleotide_archive_mcp.	Free-OSS	Web-verified ✓ Apache-2.0	Install→MCP	🖥️ query · 📄 pulls

● Compute & Pipelines

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Python · scanpy/squidpy	scanpy/squidpy/anndata for sc & spatial; pandas/numpy/scipy — all verified installed. ↳ Bash + Python 3.12; Claude writes & runs the analysis now.	Free-OSS	Known-SW	Auto · Bash	🖥️ → 📄
● Jupyter	Interactive notebooks — exploratory analysis & figures. ↳ <code>pip install jupyterlab ;</code> Claude drives notebooks via Bash.	Free-OSS	Known-SW	Install→Bash	🖥️
● R · Bioconductor	Seurat, DESeq2, limma, edgeR, survminer — the bio-stats engine. ↳ R 4.5.2 + base survival installed; add via BiocManager::install(...).	Free-OSS	Known-SW	Auto · Bash	🖥️ → 📄
● Nextflow / nf-core	Reproducible, containerised genomics pipelines at scale. ↳ Install nf-core; Claude scaffolds & launches workflows.	Free-OSS	Known-SW	Install→Bash	📄 HPC (🖥️ dev)
● Galaxy	GUI/API workflow platform for accessible bioinformatics. ↳ Galaxy public/local instance; API via browser.	Free-OSS	Known-SW	Auto · web	📄 public · 📄 local

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Snakemake	Make-style reproducible workflow engine for omics. ↳ pip/conda; Claude writes the Snakefile.	Free-OSS	Known·SW	Install→Bash	🖥️ HPC (🖥️ dev)
● cfDNA stack	Your liquid-biopsy tools: ichorCNA, MEDIPS/MeDEStrand, fragmentomics, ONT. ↳ Local conda envs; Claude orchestrates RENOVATE/MIMESIS pipelines.	Free-OSS	Known·SW	Auto·Bash	🖥️ HPC
● ChatSpatial	Spatial-transcriptomics agent — ~60 methods (Xenium/Visium/MERFISH). ↳ MCP: cafferychen777/ChatSpatial.	Free-OSS	Web-verified ✓ MIT	Install→MCP	🖥️ → 🖥️
● SCMCP	scRNA-seq analysis via scverse (scanpy). ↳ MCP: scmcphub/scmcp.	Free-OSS	Web-verified ✓ BSD-3	Install→MCP	🖥️ → 🖥️
● sc-RNA-QC skill	Official single-cell-rna-qc skill (marketplace). ↳ Install: /plugin install single-cell-rna-qc@life-sciences.	Free-OSS	Web-verified ✓ (in marketplace)	Install→Skill	🖥️
● nf-core dev skill	Official nextflow-development skill (marketplace). ↳ Install via anthropics/life-sciences marketplace.	Free-OSS	Web-verified ✓ (in marketplace)	Install→Skill	🖥️ author · 🖥️ run
● scvi-tools	Probabilistic single-cell models skill (marketplace). ↳ Install via anthropics/life-sciences marketplace.	Free-OSS	Web-verified ✓ (in marketplace)	Install→Skill	🖥️ GPU

● Statistics & Trial Design

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● statsmodels / scipy	Regression, GLM, mixed models, hypothesis tests in Python. ↳ Bash + Python; Claude writes the analysis.	Free·OSS	Known·SW	Auto·Bash	🖥️ · 🖥️
● survival · lifelines	Kaplan–Meier, Cox PH, competing risks — your endpoint engine. ↳ lifelines (Py) / survival+survminer (R) via Bash.	Free·OSS	Known·SW	Auto·Bash	🖥️ · 🖥️
● reporting standards	CONSORT / STROBE / PRISMA checklists baked into writing & review. ↳ Skills: scientific-writing / peer-review.	Free·skill	Seen·session	Skill	🖥️
● peer-review	Systematic methodology, stats, design & reproducibility review. ↳ Skill: claude-scientific-writer:peer-review.	Free·skill	Seen·session	Skill	🖥️
● critical-thinking	Rigor, bias, confounding, GRADE / Cochrane RoB appraisal. ↳ Skill: claude-scientific-writer:scientific-critical-thinking.	Free·skill	Seen·session	Skill	🖥️
◎ Bayesian · Stan/PyMC	Bayesian inference, hierarchical models, adaptive-trial priors. ↳ Install PyMC/cmdstanpy; Claude builds the model.	Free·OSS	Known·SW	Install→Bash	🖥️ → 🖥️

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Power & sample size	Power, MDE, sample-size & simulation for trial design. ↳ statsmodels.power / R pwr & simr.	Free·OSS	Known·SW	Auto·Bash	
●⊕ ClinTrials.gov MCP	Trial-registry queries over MCP (CTG API v2 + NCI CTS). ↳ MCP: cyanheads/clinicaltrialsgov-mcp-server.	Free·OSS	Web-verified ✓ Apache-2.0	Install·MCP	

Clinical & Regulatory

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● clinical-reports	Case reports (CARE), CSR (ICH-E3), SOAP, discharge summaries. ↳ Skill: claude-scientific-writer:clinical-reports.	Free·skill	Seen·session	Skill	
●⊕ decision-support	Biomarker-stratified cohorts, GRADE, treatment algorithms. ↳ Skill: claude-scientific-writer:clinical-decision-support.	Free·skill	Seen·session	Skill	
● treatment-plans	Focused LaTeX/PDF treatment plans across specialties. ↳ Skill: claude-scientific-writer:treatment-plans.	Free·skill	Seen·session	Skill	
● ClinicalTrials.gov	Global trial registry — design, comparators, recruitment. ↳ CTG API v2 via browser/HTTP.	Free·public	Known·DB	Auto·web	

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● EU CTR / CTIS	EU clinical trials register & CTIS — the EU/IT regulatory side. ↳ CTIS public portal via browser.	Free·public	Known·DB	Auto·web	
● FDA / EMA	Approvals, labels, guidance, safety — regulatory grounding. ↳ openFDA API; EMA via browser.	Free·public	Known·DB	Auto·web	
● WHO ICTRP	Cross-registry trial search (incl. non-US/EU). ↳ ICTRP portal via browser.	Free·public	Known·DB	Auto·web	

● Visualization & Figures

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● BioRender	Publication figures from a vast bio icon/template library. ↳ MCP connected now — search icons & templates.	Freemium (pub export paid)	Seen·session	Auto·MCP	
● Mermaid	Flowcharts, sequence & ER diagrams from text; live render. ↳ MCP connected now — validate & render diagrams.	Free·OSS	Seen·session	Auto·MCP	
● Figma	Vector design for posters, schematics, slide polish. ↳ MCP connected now (auth) — pull frames & assets.	Freemium	Seen·session	Auto·MCP	

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● scientific-schematics	Pathways, architectures, mechanism diagrams (Nano Banana Pro). ↳ Skill: claude-scientific-writer:scientific-schematics.	Free-skill (image API may bill)	Seen-session	Skill	
● infographics	Data-driven infographics with colourblind-safe palettes. ↳ Skill: claude-scientific-writer:infographics.	Free-skill (image API may bill)	Seen-session	Skill	
● image-gen · Nano/FLUX	General images, illustrations, photoreal assets (FLUX/Gemini). ↳ Skill: claude-scientific-writer:generate-image.	Free-skill (FLUX/Gemini bills)	Seen-session	Skill	
● gpt-image-2 (hero)	SOTA hero/cover artwork with reliable in-image text. ↳ Manual ChatGPT handoff (Path A) — not wired to API.	Paid/lic. (OpenAI / ChatGPT)	Known (OpenAI)	Manual	

● Writing & Publishing

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● scientific-writing	IMRAD prose (never bullets), citations, reporting guidelines. ↳ Skill: claude-scientific-writer:scientific-writing.	Free-skill	Seen-session	Skill	
● citation-management	Scholar/PubMed metadata, DOI→BibTeX, reference validation. ↳ Skill: claude-scientific-writer:citation-management.	Free-skill	Seen-session	Skill	

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● venue-templates	Nature/Science/PLOS/IEEE + NIH/NSF formatting & rules. ↳ Skill: claude-scientific-writer:venue-templates.	Free-skill	Seen-session	Skill	
● Office docs	Author & edit docx / pptx / xlsx / pdf — tracked changes, slides, sheets. ↳ Skills: claude-scientific-writer:docx/pptx/xlsx/pdf.	Free-skill	Seen-session	Skill	
● markdown	Convert PDF/DOCX/PPTX/audio/HTML → Markdown for ingest. ↳ Skill: claude-scientific-writer:markdown.	Free-skill	Seen-session	Skill	
● humanizer	Strip AI-tells; enforce natural prose (your house style). ↳ Skill: humanizer.	Free-skill	Seen-session	Skill	
● preparing-slides	Your Zoppoli house-style .pptx engine (DIMI/GOIRC/...). ↳ Skill: preparing-slides — your venues, palettes, rules.	Free-skill	Seen-session	Skill	
● research-grants	NSF/NIH/DOE/DARPA + EU proposals, significance, budgets. ↳ Skill: claude-scientific-writer:research-grants.	Free-skill	Seen-session	Skill	
● paper-2-web	Paper → website / video / poster for dissemination. ↳ Skill: claude-scientific-writer:paper-2-web.	Free-skill (video may bill)	Seen-session	Skill	 (video heavier)
● posters (LaTeX/PPTX)	Conference posters via beamerposter/tikz or PPTX. ↳ Skills: latex-posters / pptx-posters.	Free-skill	Seen-session	Skill	 (local LaTeX)

Orchestration & Lab Ops

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Google Drive	Cloud store — data, manuscripts, shared docs; read & write. ↳, MCP connected now — search, read, create files.	Freemium	Seen-session	Auto·MCP	
● Gmail	Search threads, draft, label — collaborator & journal comms. ↳, MCP connected now — search & draft (you send).	Freemium	Seen-session	Auto·MCP	
● Google Calendar	Schedule meetings, deadlines, find slots. ↳, MCP connected now — list, create, suggest times.	Freemium	Seen-session	Auto·MCP	
● Slack	Team/consortium channels — search & post. ↳, MCP connected (auth) — search & message.	Freemium	Seen-session	Auto·MCP	
● Workflows / subagents	Deterministic multi-agent fan-out: review, migrate, research. ↳, Workflow tool + Agent subagents — orchestrate at scale.	Free-built-in (uses tokens)	Seen-session	Built-in	 (more tokens at scale)
● schedule / cron	Recurring remote agents — overnight pipelines, polling, digests. ↳, Skill: schedule — cron-style routines & one-offs.	Free-skill	Seen-session	Skill	 (cloud-run)

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
● Codex · GPT-5.5	Second engine — decorrelated review, long shell agents, red-team. ↳ codex exec via Bash; Skill: codex-dispatch.	Paid/lic. (ChatGPT/OpenAI)	Configured (CLAUDE.md)	Auto · Bash	🖥️ (codex CLI)
●🕒 Claude for Life Sciences	Anthropic's official life-sci plugin marketplace — remote MCPs + skills. ↳ Install: /plugin marketplace add anthropics/life-sciences.	Free (some servers it ships are paid)	Web-verified ✓ (412★)	Install (channel)	🖥️
●🕒 BioContextAI Registry	Nature-Biotech-backed index of ~60 biomedical MCP servers. ↳ Browse biocontext.ai/registry · github.com/biocontext-ai/registry.	Free-OSS	Web-verified ✓ repo	Discovery	🖥️
●🕒 BioContextAI meta-MCP	Meta layer: knowledgebase / skill-to-mcp / protocol / registry servers. ↳ github.com/biocontext-ai — add the meta server to your config.	Free-OSS	Web-verified ✓ (biocontext-ai)	Install-MCP	🖥️
●🕒 NAR DB Collection	2,173 curated molecular-biology databases (2026 issue) — the cross-check. ↳ Reference index: academic.oup.com/nar.	Free-public	Known-DB	Discovery	🖥️

Station	Functions · how to connect	Cost	Real?	Invocation	Runs best on
☉ BioMCP	One server federating literature, variants, trials & cancer genomics. ↳ MCP: genomoncology/biomcp — the single best biomedical addition.	Free-OSS	Web-verified ✓ MIT	Install→MCP	🖥️ (wraps remote APIs)

☉ Interchange lines — the multi-domain servers that connect the lines

Each is one MCP server that spans several lines; wiring it lights up every station it touches.

BioMCP — literature · genomic · trials federation

- **Connect:** MCP: genomoncology/biomcp — the single best biomedical addition.
- **Functions:** One server federating literature, variants, trials & cancer genomics.
- **Invocation:** Install→MCP · **Runs best on:** 🖥️ (wraps remote APIs)
- **Stops it serves:** BioMCP · DepMap / OncoKB · cBioPortal · ClinVar / gnomAD · PubMed · ClinTrials.gov MCP

Open Targets — target · disease · treatment

- **Connect:** Official remote MCP: mcp.platform.opentargets.org/mcp.
- **Functions:** Target–disease–drug associations, genetics, tractability.
- **Invocation:** Install→MCP · **Runs best on:** 🖥️ (remote)
- **Stops it serves:** Open Targets MCP · cBioPortal MCP · decision-support

Discovery ring — Life-Sciences marketplace + BioContextAI

- **Connect:** /plugin marketplace add anthropics/life-sciences (marketplace) and browse biocontext.ai/registry (registry). The install/discovery backbone.
- **Invocation:** Install (channel) + Discovery · **Runs best on:** 🖥️ any laptop
- **Stops it serves:** Claude for Life Sciences · BioContextAI Registry · BioContextAI meta-MCP · NAR DB Collection

The hub & the second engine

- 🟡 **Claude Code (central interchange)** — your workstation. Every line meets here: Claude routes the work, reasons, writes & runs code, and dispatches the rest. **Invocation:** this is the station you sit in. **Runs best on:** 🖥️ workstation (it drives everything else, local + cloud).

-  **Codex · GPT-5.5 (sister terminus)** — the second engine for decorrelated review, long shell-agent runs and red-teaming. **Connect:** `codex exec --skip-git-repo-check "..."` via Bash, or the `codex-dispatch` skill. **Invocation:** `Auto·Bash` / `Skill`. **Runs best on:**  workstation (where the codex CLI + ChatGPT auth live).

At a glance — where the work happens

-  **Any laptop (most of the map):** all literature, clinical & regulatory lines, every writing/visualization skill, the trial-registry MCPs, and the orchestration connectors (Drive/Gmail/Calendar/Slack/workflows/schedule). If it's a cloud connector, a remote MCP, a public API or a skill, the heavy lifting happens off your machine — you only need Claude Code + internet.
-  **Workstation:** anything touching your local envs, credentials or data — Synapse controlled-access pulls, Jupyter, light single-cell (scanpy/Seurat) exploration, the Codex CLI. Also the natural home for the MCP servers that *execute* analysis locally (ChatSpatial, SCMCP) when datasets are modest.
-  **HPC:** the genomics compute spine — Nextflow/nf-core & Snakemake pipelines, the cfDNA/liquid-biopsy stack (RENOVATE / MIMESIS), large single-cell & spatial atlases, GPU model training (scvi-tools), and bulk SRA/GDC/ENA data pulls. Develop on the workstation, run at scale on HPC.

Generated by `make_guide.py` from `generate_map.py`. Edit the map data or the `INVOKE` / `RUNS_ON` dicts and re-run to refresh.